

Methods	Year	Parameters	Dice	iou	Input size
TransNetR (4)	2023	27.27	87.06	80.16	?
SegMed-Net (8)	2022	59.9	90.8	83.70	352*352
PVT-Cascade (3)	2023	35.2	91.1	86.3	352*352
MegaNet (2)	2024	44.1	91.3	86.3	352*352
Fu-TransHNet (5)	2023	?	91.6	86.6	352*352
LiteBoundD (1)	2026	32.8	91.9	84.1	352*352
AAPCNet (7)	2025	38.2	92.10	86.9	256*256
DVFIP-Net	2026	26.54	02.29	04.78	352*352
SAM2-FNet	2026	2.72	04.29	87.40	352*352
MLB-Net	2026	87.7	92.60	87.80	352*352
MSPN	2026	28.56	92.70	87.90	352*352
HCA-Net	2026	89.70	93.60	89.7	352*352
Ours	2026	38.5	93.89	88.66	352*352
PolyMamba-Net (6)	2026	25.3	94.2	89.1	352*352

1. Sharpening Lightweight Models for Generalized Polyp Segmentation: A Boundary Guided Distillation from Foundation Models
2. “Meganet: Multi-scale edge-guided attention network for weak boundary polyp segmentation
3. “Medical image segmentation via cascaded attention decoding
4. TransNetR: Transformer-based Residual Network for Polyp Segmentation with Multi-Center Out-of-Distribution Testin
5. Cooperation Learning Enhanced Colonic Polyp Segmentation Based on TransformerCNN Fusion
6. PolyMamba-Net: a lightweight and boundary-aware network for real-time polyp segmentation in colonoscopy
7. Attention-Guided Asymmetric Multiscale Polyp Segmentation Network
8. SegMed-Net: a domain-aware dual encoder-decoder network with dense feature fusion for enhanced medical image segmentation

Boundary-Frequency Injection Module:

- Decomposing frequency components
 - Frequency Feature Decomposer (FFD)
 - Spilt skip features into high-frequency and low-frequency using 2D FFT
 - * High-frequency -> Boundaries, edges
 - * Low-frequency -> semantic content
- Using boundary information for better feature guidance
 - Boundary-guided Squeeze-Excitation Injection (BSEI)
 - Uses Sobel operators to detect boundaries in skip features
 - Creates spatial attention maps from boundary information
 - Uses cross-attention between decoder features and boundary-weighted encoder features
- Adaptively fusing these different feature representations
 - Multi-scale Frequency Adaptive Fusion (MFAF)
 - Learns soft weights to combine high-frequency, low-frequency, and boundary-enhanced features

- Uses channel-wise gating to decide the importance of each component

Purpose:

The BFIM replaces standard skip connections to:

- **Better preserve boundary information** during up-sampling
- **Adaptively emphasize** important frequency components
- **Improve segmentation accuracy** especially for object boundaries
- **Provide learnable fusion** instead of simple concatenation

Models	Seen				Unseen						Epoch
	Kvasir-Seg		ClinicDB		CVC-300		ColonDB		ETIS		
	Dice	iou	Dice	iou	Dice	iou	Dice	iou	Dice	iou	
HCA-Net	93.6	89.7	95.6	90.9	91.2	87.3	83.4	76.9	83.3	78.9	
Ours	92.40	86.08	95.19	90.84	91.49	84.32	81.26	70.44	87.39	77.61	1370
Ours-2	92.58	86.41	95.31	91.04			81.45	70.95			1405
	92.03		95.27	90.97			83.34	73.75			

Models	Seen				Unseen						Epoch
	Kvasir-Seg		ClinicDB		CVC-300		ColonDB		ETIS		
	Dice	iou	Dice	iou	Dice	iou	Dice	iou	Dice	iou	
HCA-Net	93.6	89.7	95.6	90.9	91.2	87.3	83.4	76.9	83.3	78.9	
	93.49	87.77	95.31	91.04	91.61	84.52	83.34	73.75	88.69	79.67	
Our	93.49	87.77	94.92	90.33	90.33	82.52	79.72	69.16	88.69	79.67	100
Ours	92.58	86.41	95.31	91.04	91.49	84.32	83.34	73.75	87.39	77.61	1370
Ours-93.04	03.29				52.19	19.38			84.96	73.85	
Ours-93.66	92.75	86.48	95.20	90.83	91.61	84.52	82.81	72.96	85.39	74.50	2836

Introduction

Colorectal polyps are abnormal tissue growths that may develop into malignant lesions if they are not detected and removed at an early stage. Colonoscopy is routinely employed for examination of the lower gastrointestinal tract and enables clinicians to identify and resect suspicious polyps. However, the reliability of visual inspection is affected by operator experience, fatigue, imaging conditions, and the heterogeneous appearance of colorectal lesions. Computer-aided polyp segmentation can assist clinical examination by providing pixel-level localization of suspected lesions and by presenting their spatial extent and boundaries more explicitly.

Accurate polyp segmentation remains challenging because colorectal lesions exhibit substantial variations in size, morphology, texture, color, and location. Some polyps occupy a large and visually distinctive region, whereas others are small, flat, or partially concealed by intestinal folds. In addition, visual similarity between lesion tissue and the surrounding mucosa frequently produces weak or ambiguous boundaries. Specular reflections, motion blur, bubbles, shadows, fecal material, and medical instruments may further interfere with feature extraction. These factors make it difficult for a segmentation model to obtain reliable semantic localization and precise boundary delineation simultaneously.

U-shaped convolutional architectures have been extensively used in medical image segmentation because their contracting paths extract hierarchical semantic features, while their expanding paths progressively recover spatial resolution. Skip connections transfer relatively high-resolution encoder representations to the decoder and therefore facilitate lesion localization. Nevertheless, conventional skip connections may also introduce redundant texture responses and background interference because encoder features are transferred without explicit feature selection. Moreover, repeated spatial down-sampling may remove subtle structural information required for segmenting small or weakly bounded polyps. U-Net established the fundamental contracting-expanding design, while subsequent architectures, including UNet++, redesigned skip pathways and incorporated deep supervision to reduce the semantic discrepancy between encoder and decoder representations [1,2].

To improve global-context modeling, transformer-based and hybrid convolution-transformer architectures have been investigated for medical image segmentation. TransUNet combines convolutional feature extraction with transformer encoding to capture long-range dependencies, while Polyp-PVT employs hierarchical transformer representations and cross-level feature aggregation for polyp segmentation [3,4]. These methods demonstrate the value of global contextual information; however, transformer-based designs can introduce considerable computational complexity and may still require dedicated mechanisms for recovering local boundary information.

Modern convolutional architectures provide an alternative means of obtaining strong hierarchical representations without relying on self-attention. ConvNeXt modernizes conventional convolutional design through large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth. These characteristics enable ConvNeXt to retain the spatial inductive biases and computational regularity of convolutional networks while providing competitive representation capacity [5].

Despite these advances, three interconnected limitations remain important for polyp segmentation. First, the large variation in lesion size requires contextual representations with multiple receptive fields. Second, weak boundaries require a mechanism that selectively enhances local structural variations rather than indiscriminately transferring skip features. Third, deep semantic representations must be combined with shallow spatial information without allowing noisy low-level responses to dominate the final prediction. Addressing these limitations within a computationally moderate architecture is particularly important for applications in which both accuracy and practical inference efficiency are required.

This study proposes a multi-scale and boundary-sensitive ConvNeXt-UNet for automatic colorectal polyp segmentation. The model employs a ConvNeXt-Tiny encoder to generate four levels of hierarchical representations. A lightweight residual bottleneck and a Multi-Scale Context module are applied to the deepest encoder feature to aggregate local, enlarged-receptive-field, and global information. During decoding, a Boundary-Sensitive Enhancement and Integration module refines the concatenated encoder and decoder features by learning spatial edge-response attention. In parallel, a shallow Detail Branch preserves high-resolution appearance information directly from the input image. A Gated Detail Fusion module selectively injects this detailed representation into the final decoder stage. Three auxiliary output heads provide intermediate supervision during training. The complete architecture contains approximately 29.61 million trainable parameters.

The principal contributions of this work are summarized as follows:

1. A ConvNeXt-Tiny-based U-shaped segmentation architecture is developed to combine hierarchical convolutional feature extraction with a lightweight decoder for colorectal polyp segmentation.
2. A Multi-Scale Context module is introduced at the bottleneck to aggregate features obtained using three dilated depthwise convolution branches and a global-context branch.
3. A Boundary-Sensitive Enhancement and Integration module is designed to refine skip-connected features through learned spatial edge-response modulation.
4. A high-resolution Detail Branch and Gated Detail Fusion module are introduced to preserve shallow spatial information and selectively combine it with the final decoder representation.

5. Intermediate prediction heads are incorporated to support deep supervision and improve optimization of multi-level decoder features.
6. The proposed model is evaluated on Kvasir-SEG, CVC-ClinicDB, CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB using in-distribution and cross-dataset testing protocols.

Related work

Convolutional encoder-decoder architectures

Fully convolutional encoder-decoder networks form the foundation of modern medical image segmentation. U-Net introduced a symmetric contracting and expanding architecture in which high-resolution encoder representations are transferred to the decoder through skip connections. This design enables the network to combine contextual representations with spatially detailed features and has become a standard baseline for biomedical segmentation [1].

Subsequent studies modified the U-shaped architecture to improve feature fusion and optimization. UNet++ introduced nested dense skip pathways to reduce the semantic discrepancy between encoder and decoder features and employed deep supervision at multiple decoder depths [2]. Such developments demonstrated that the design of skip connections is as important as the choice of encoder because direct feature transfer may combine representations with substantially different semantic characteristics.

In polyp segmentation, computationally efficient convolutional networks have also received considerable attention. HarDNet-MSEG combines a low-memory-traffic convolutional backbone with a lightweight decoder and illustrates that competitive segmentation performance can be obtained without a computationally expensive transformer architecture [6]. However, standard convolutional models generally require additional mechanisms to capture broad contextual relationships and refine weak lesion boundaries.

PraNet introduced a parallel partial decoder to aggregate high-level features and a reverse-attention strategy to progressively refine uncertain polyp regions [7]. Its recurrent interaction between regional predictions and boundary cues demonstrated the effectiveness of explicitly directing the decoder toward inadequately segmented regions. Nevertheless, reverse-attention refinement depends on preceding prediction maps and does not directly solve the problem of noisy information transferred through conventional skip connections.

Global and multi-scale contextual modeling

The considerable variation in polyp size motivates the use of multi-scale contextual modeling. A segmentation network must preserve local details for small lesions while maintaining a sufficiently large receptive field to identify extensive or irregularly shaped polyps. Dilated convolutions, pyramid pooling, multi-branch receptive fields, and cross-scale feature aggregation have consequently been incorporated into many segmentation architectures.

Transformer-based approaches address this problem by modeling long-range spatial relationships. TransUNet uses a transformer encoder over convolutional feature representations to capture global context while retaining high-resolution CNN features for localization [3]. Polyp-PVT employs a pyramid vision transformer and specialized fusion modules to integrate semantic and positional information across hierarchical feature levels [4]. These methods improve long-range feature interaction, but their attention operations and cross-level fusion mechanisms may increase architectural and computational complexity.

Recent contextual-information-flow and multi-stage architectures similarly emphasize the need to coordinate local lesion appearance with broader anatomical context. State-space and frequency-oriented architectures have also been investigated. Polyp-Mamba combines multi-frequency perception with a gated selection mechanism to model spatial dependencies and frequency-sensitive representations [8]. The present work does not employ frequency decomposition or a state-space operator; instead, it uses parallel dilated depthwise convolutions and global average pooling to obtain multi-scale contextual information within a purely convolutional framework.

Boundary-aware polyp segmentation

Weak and visually ambiguous lesion boundaries are among the principal difficulties in colorectal polyp segmentation. Boundary-aware networks attempt to preserve or reconstruct structural information that may disappear during repeated down-sampling. These methods commonly introduce explicit edge supervision, auxiliary contour branches, gradient-derived features, reverse attention, or boundary-sensitive feature fusion.

MEGANet employs multi-scale edge-guided attention and incorporates high-frequency information derived from image pyramids to improve segmentation of weak boundaries [9]. Its findings demonstrate that explicitly retaining edge-related information can reduce deterioration of boundary features in deep networks. MSBP-Net performs multi-scale boundary prediction to improve contour delineation while maintaining relatively low model complexity [10]. This line of research indicates that boundary information should be modeled at more than one scale because polyp contours may contain both broad structural transitions and small irregularities.

Explicit edge extraction may also respond to irrelevant structures such as folds, reflections, bubbles, and instruments. Furthermore, additional boundary labels or edge-generation procedures may increase the complexity of the training pipeline. The BSEI module in the proposed model follows a different strategy. It derives a learned spatial response from transformed skip features and uses this response to modulate the features through residual enhancement. Therefore, the method is boundary-sensitive but does not depend on Sobel filtering, Laplacian pyramids, explicit contour labels, or a separate boundary loss.

Detail preservation and adaptive feature fusion

Deep semantic features provide strong category discrimination but have reduced spatial resolution. Conversely, shallow features preserve local appearance and edge information but may contain substantial background noise. Effective segmentation therefore requires controlled interaction between semantic and detailed representations.

Conventional U-shaped networks concatenate or add shallow and deep features directly. Although these operations are computationally simple, they assign no explicit relevance to detailed information. Attention and gating mechanisms provide a more selective alternative by estimating the contribution of one representation conditioned on another. Recent polyp-segmentation architectures employ specialized cross-scale or attention-based fusion to suppress irrelevant responses and improve lesion localization.

The proposed method addresses detail preservation using a shallow Detail Branch operating at one-half of the input resolution. Instead of directly adding this feature to the decoder, the Gated Detail Fusion module estimates a gate from the concatenated decoder and detail representations. The gate modulates the detail feature before final fusion, enabling the network to retain high-resolution information while limiting transfer of irrelevant low-level responses.

Positioning of the proposed method

Existing approaches have improved polyp segmentation through reverse attention, transformer-based contextual modeling, frequency processing, explicit boundary prediction, or complex multi-stage refinement. Nevertheless, many methods address context, boundaries, and fine details as separate problems. The proposed architecture integrates these requirements within a unified convolutional encoder-decoder framework.

Unlike transformer-based methods, the model employs a ConvNeXt-Tiny encoder and depthwise separable convolutions. Unlike frequency-oriented networks, it does not require Fourier or wavelet decomposition. Unlike explicitly supervised boundary networks, it derives a learned local-response map from intermediate features. Finally, unlike direct shallow-deep fusion, it uses a conditioned gate to control the contribution of the high-resolution detail representation. These design choices provide a coherent framework for investigating whether multi-scale context, boundary-sensitive skip refinement, and selective detail fusion can jointly improve polyp segmentation and cross-dataset generalization.

Method

Overall Architecture

The proposed framework follows a U-shaped encoder–decoder architecture built upon a ConvNeXt-Tiny encoder. The central objective is to exploit frequency information extracted from deep semantic features to adaptively regulate multi-scale feature processing during decoding. Rather than independently applying frequency enhancement and multi-scale convolution, the proposed model establishes a cross-level interaction between the bottleneck and the first decoder stage.

Let

$$I \in \mathbb{R}^{3 \times H \times W}$$

denote an input colonoscopy image. The ConvNeXt-Tiny encoder extracts four hierarchical feature maps,

$$\{F_1, F_2, F_3, F_4\} = E(I),$$

where

$$F_1 \in \mathbb{R}^{96 \times H/4 \times W/4}, \quad F_2 \in \mathbb{R}^{192 \times H/8 \times W/8},$$

$$F_3 \in \mathbb{R}^{384 \times H/16 \times W/16}, \quad F_4 \in \mathbb{R}^{768 \times H/32 \times W/32}.$$

For an input resolution of 352×352 , their spatial dimensions are 88×88 , 44×44 , 22×22 , and 11×11 , respectively.

The deepest encoder feature F_4 is first processed by a Frequency-Aware Feature Enhancement Module (FAFEM), which produces an enhanced bottleneck representation together with a compact frequency descriptor. The enhanced representation is passed through a lightweight residual bottleneck, whereas the frequency descriptor is propagated to the first decoder stage. After the decoder feature is fused with F_3 , a Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses the frequency descriptor to adaptively regulate its multi-scale convolution branches.

The overall processing can therefore be summarized as

$$(\tilde{F}_4, \mathbf{z}_f) = \mathcal{F}_{\text{FAFEM}}(F_4),$$

$$B = \mathcal{B}(\tilde{F}_4),$$

$$D_4 = \mathcal{M}_{\text{RFG}}(\mathcal{S}_3[\mathcal{D}_4(B), F_3], \mathbf{z}_f),$$

followed by

$$D_3 = \mathcal{S}_2[\mathcal{D}_3(D_4), F_2],$$

$$D_2 = \mathcal{S}_1[\mathcal{D}_2(D_3), F_1],$$

$$D_1 = \mathcal{D}_1(D_2).$$

Here, $\mathcal{D}_i$ denotes a decoder block, $\mathcal{S}_i$ denotes the corresponding skip-feature fusion operation, $\mathbf{z} * f$ is the frequency descriptor extracted at the bottleneck, and $\mathcal{M}^*\text{RFG}$ denotes the proposed residual frequency-guided multi-scale refinement mechanism.

The final prediction head transforms D_1 into a single-channel segmentation logit map, which is bilinearly resized to the original image resolution.

ConvNeXt-Tiny Encoder

The encoder employs ConvNeXt-Tiny with stage depths

$$[3, 3, 9, 3]$$

and channel dimensions

$$[96, 192, 384, 768].$$

Before entering the encoder, each RGB image is normalized using the ImageNet statistics

$$\boldsymbol{\mu} = (0.485, 0.456, 0.406), \quad \boldsymbol{\sigma} = (0.229, 0.224, 0.225).$$

The first stage uses a 4×4 convolution with stride 4 to generate F_1 . Each subsequent stage begins with spatial downsampling using a 2×2 stride-2 convolution.

For an input feature X , a ConvNeXt block first performs a 7×7 depthwise convolution. The resulting feature is normalized in channels-last format and processed by a pointwise multilayer perceptron with a fourfold channel expansion:

$$Z = \text{DWConv}_{7 \times 7}(X),$$

$$V = W_2(\text{GELU}(W_1(\text{LN}(Z)))).$$

The block output is obtained through layer-scaled residual learning,

$$Y = X + \text{DropPath}(\gamma \odot V),$$

where γ is a learnable layer-scale parameter initialized to 10^{-6} .

A stochastic-depth rate of 0.25 is used throughout the encoder. Feature dropout with probability 0.2 is additionally applied after the second, third, and fourth encoder stages.

Frequency-Aware Bottleneck Enhancement

The deepest encoder representation F_4 contains strong semantic information but has undergone substantial spatial downsampling. To expose complementary frequency characteristics of this representation, FAFEM is applied exclusively at the bottleneck.

Given

$$X = F_4,$$

a local low-frequency approximation is obtained using 3×3 average pooling,

$$X_L = \text{AvgPool}_{3 \times 3}(X),$$

and the corresponding high-frequency component is defined as

$$X_H = X - X_L.$$

The low- and high-frequency components are independently transformed using depthwise convolutional branches,

$$\hat{X}_L = \phi_L(X_L), \quad \hat{X}_H = \phi_H(X_H),$$

where each $\phi(\cdot)$ contains a 3×3 depthwise convolution followed by GELU activation.

The two components are concatenated and globally pooled to obtain the frequency descriptor

$$\mathbf{z}_f = \text{GAP}([\hat{X}_L, \hat{X}_H]) \in \mathbb{R}^{2C}.$$

Since $C = 768$ at the bottleneck,

$$\mathbf{z}_f \in \mathbb{R}^{1536}.$$

A lightweight channel-gating network generates adaptive weights for the low- and high-frequency representations. Their weighted combination is spatially refined and introduced through a residual connection:

$$\tilde{F}_4 = F_4 + \gamma_f \mathcal{R}_f(w_L \odot \hat{X}_L + w_H \odot \hat{X}_H),$$

where γ_f is a learnable scalar initialized to zero.

Importantly, FAFEM itself is not introduced in this work. In the proposed architecture, its frequency descriptor $\mathbf{z}_f$ is additionally reused to condition multi-scale processing at a later decoder stage.

Lightweight Residual Bottleneck

The frequency-enhanced representation $\tilde{F}_4$ is subsequently processed by a lightweight residual bottleneck.

A 1×1 convolution first reduces the number of channels from 768 to 192:

$$X_r = \text{GELU}\left(\text{LN}\left(\text{Conv}_{1 \times 1}(\tilde{F}_4)\right)\right).$$

A 3×3 depthwise-separable convolution performs spatial transformation in the reduced feature space. The representation is subsequently projected back to 768 channels.

The bottleneck output is formulated as

$$B = \tilde{F}_4 + \gamma_b \mathcal{H}(\tilde{F}_4),$$

where $\mathcal{H}(\cdot)$ denotes the bottleneck transformation and γ_b is a learnable scalar initialized to zero.

This initialization makes the bottleneck behave initially as an identity mapping and allows its transformation to be progressively introduced during optimization.

Residual Frequency-Guided Multi-Scale Convolution

The principal architectural contribution of this work is the Residual Frequency-Guided Multi-Scale Convolution mechanism.

After the bottleneck feature is upsampled from $H/32$ to $H/16$, the decoder representation is reduced from 768 to 384 channels and concatenated with the Stage-3 encoder feature F_3 . A lightweight fusion block transforms the resulting 768-channel representation into

$$X_3 \in \mathbb{R}^{384 \times H/16 \times W/16}.$$

For a 352×352 input,

$$X_3 \in \mathbb{R}^{384 \times 22 \times 22}.$$

A conventional lightweight multi-scale convolution block first expands the feature dimension from C to $2C$:

$$X_e = \text{ReLU6}(\text{BN}(\text{Conv}_{1 \times 1}(X_3))).$$

Three parallel depthwise convolution branches with kernel sizes

$$k \in \{1, 3, 5\}$$

then generate multi-scale responses,

$$B_k = \phi_k(X_e),$$

where each ϕ_k consists of a depthwise convolution, Batch Normalization, and ReLU6 activation.

In a conventional MSCB, these branches are combined uniformly:

$$M_{\text{ord}} = B_1 + B_3 + B_5.$$

Instead of treating the three receptive fields identically for every image, the proposed mechanism conditions their relative contributions on the bottleneck frequency descriptor $\mathbf{z}_f$.

A lightweight guidance network maps the 1536-dimensional descriptor to three branch logits:

$$\mathbf{q} = W_{g2}(\text{GELU}(W_{g1}\mathbf{z}_f)),$$

where

$$W_{g1} : 1536 \rightarrow 48, \quad W_{g2} : 48 \rightarrow 3.$$

The normalized branch weights are calculated as

$$[\alpha_1, \alpha_3, \alpha_5] = 3 \text{Softmax}(\mathbf{q}),$$

such that

$$\alpha_1 + \alpha_3 + \alpha_5 = 3.$$

The frequency-guided response becomes

$$M_{\text{guide}} = \alpha_1 B_1 + \alpha_3 B_3 + \alpha_5 B_5.$$

Directly replacing the conventional multi-scale response with the guided response may excessively perturb an already effective MSCB representation. We therefore introduce the guidance as a bounded residual deviation:

$$M_{\text{RFG}} = M_{\text{ord}} + s(M_{\text{guide}} - M_{\text{ord}}),$$

where

$$0 \leq s \leq 0.5.$$

The learnable guidance-strength parameter s is initialized to 0.05. Consequently, the frequency-conditioned branch regulation is introduced conservatively during early optimization.

Furthermore, the final guidance-layer weights and biases are initialized such that the initial branch logits are equal. Therefore,

$$\alpha_1 = \alpha_3 = \alpha_5 = 1$$

at initialization, which gives

$$M_{\text{guide}} = M_{\text{ord}}.$$

The module thus initially reproduces the conventional multi-scale path, while optimization is allowed to learn image-dependent deviations from uniform branch weighting.

The resulting feature is processed using channel shuffle and a 1×1 projection from $2C$ back to C , followed by Batch Normalization. A final residual connection preserves the input representation:

$$Y_3 = X_3 + \mathcal{P}(\text{Shuffle}(M_{\text{RFG}})).$$

This formulation establishes an explicit cross-level connection between deep frequency information extracted at the bottleneck and multi-scale spatial refinement performed at $H/16$ resolution.

Progressive Decoder and Skip Fusion

After frequency-guided Stage-3 refinement, the decoder progressively restores spatial resolution using bilinear upsampling and depthwise-separable convolution.

The first refined decoder feature has 384 channels:

$$D_4 = Y_3.$$

The subsequent decoder stages are

$$D_3 = \mathcal{S}_2[\mathcal{D}_3(D_4), F_2],$$

$$D_2 = \mathcal{S}_1[\mathcal{D}_2(D_3), F_1],$$

followed by

$$D_1 = \mathcal{D}_1(D_2).$$

Their dimensions for a 352×352 input are

$$D_4 : 384 \times 22 \times 22,$$

$$D_3 : 192 \times 44 \times 44,$$

$$D_2 : 96 \times 88 \times 88,$$

and

$$D_1 : 96 \times 176 \times 176.$$

At the F_2 and F_1 skip levels, the upsampled decoder representation is concatenated with the corresponding encoder feature and processed by a lightweight fusion block containing a 1×1 projection followed by a depthwise-separable 3×3 convolution.

No additional attention mechanism, frequency module, MSCB, or boundary-specific refinement block is applied at these decoder stages.

The final decoder stage has no direct encoder skip connection because the ConvNeXt stem begins at one-quarter of the input resolution.

Prediction Head

The final decoder feature D_1 is processed using a lightweight refinement head. Two depthwise-separable 3×3 convolutions transform the feature dimensions as

$$96 \rightarrow 96 \rightarrow 48.$$

Each transformation is followed by Layer Normalization and GELU activation. A final 1×1 convolution produces a single-channel segmentation logit map:

$$P_0 = \text{Conv}_{1 \times 1}(\mathcal{R}(D_1)).$$

The prediction is then bilinearly resized from $H/2 \times W/2$ to the original input resolution:

$$P = \text{Up}_{H,W}(P_0).$$

The network directly returns logits. During inference, the foreground probability map is obtained as

$$\hat{Y} = \sigma(P),$$

where $\sigma(\cdot)$ denotes the sigmoid function.

The selected architecture does not employ auxiliary prediction heads or deep supervision.

Model Complexity

The complete proposed model contains

30,057,415

trainable parameters, corresponding to approximately

30.06 M

parameters.

The additional frequency-guided mechanism is introduced only at the first decoder–encoder fusion stage. The remaining decoder retains lightweight depthwise-separable convolution and conventional skip fusion, limiting the architectural modification to the stage at which high-level semantic information is first combined with the relatively high-resolution F_3 representation.

Experiments

Dataset:

To verify the effectiveness of the models, we adopted five public datasets (including Kvasir-SEG [ho-32], CVC-ClinicDB [ho-33], ETIS [ho-34], CVC-ColonDB [ho-35] and CVC-300 [ho-36]) commonly used in existing polyp segmentation studies. These images are acquired from colonoscopy video sequences, and all of them contain the corresponding ground truth. For fair comparison, according to table 1, we split the datasets as in PraNet [8]. The training set consists of 900 images from the Kvasir-SEG and 550 images from the CVC-ClinicDB. The remaining samples in the Kvasir-SEG and the CVC-ClinicDB, and the other three datasets are used as five independent test sets. In this study, we randomly selected 145 images (10 % of the training set) from the training set as the validation set, serving as the criterion for saving the model during training.

Table 1 Number, resolution, and trainin/test split of the five public polyp segmentation datasets. Resized to 352*352.

Dataset	Number	Train	Test	Resolution
Kvasir-Seg	1000	900	500	500*574
CVC-ClinicDB	612	550	62	288*384
CVC-ColonDB	380	-	380	500*570
ETIS-LaribPolypDB	196	-	196	966*1225
CVC-300	60	-	60	500*570

Loss function

Evaluation metrics

To objectively evaluate model performance, two commonly used segmentation metrics are employed: the mean Intersection over Union (mIoU) and mean Dice coefficient (mDice). In addition, in order to evaluate the performance of the model more comprehensively, this paper also adds four evaluation indicators commonly used in the field of salient object detection, namely S-measure ($S\alpha$) [ho-f20], F-measure[], E-measure () [ho-fan] and mean absolute error (MAE) [ho-peraz]. The detailed definitions and formulas for these metrics are as follows: mIoU and mDice are commonly used to measure the similarity between the prediction map and the real map. They quantitatively analyze the spatial overlap between the prediction map and the real map. The higher the value, the better the segmentation accuracy of the model. Among them, the average

means that the IoU coefficient and Dice coefficient of each category are averaged, but there is only one category of polyps here. Thus, their calculation formulas are as follows:

$$mIoU = \frac{TP}{TP + FN + FP}$$

$$mDice = \frac{2 * TP}{2 * TP + FP + FN}$$

Where TP (True Positive) indicates the real sample, that is, the forecast is a positive sample, and the actual sample is also a positive sample; FP (False Positive) indicates the false positive sample, that is, the prediction is a positive sample, and the actual sample is a negative sample; FN (False Negative) indicates the false negative sample, that is, the prediction is a negative sample, and the actual sample is a positive sample.

S-measure, S_α , is used to measure the structural similarity between region perception (S_r) and object perception (S_o) between the predicted graph and the real graph. It comprehensively considers the similarity of spatial structure and regional contour, and is a comprehensive evaluation of segmentation performance. Thus, this calculation formulas are as follows:

$$S_\alpha = \alpha * S_o + (1 - \alpha) * S_r$$

Where α is the weight parameter, and the default setting is 0.5.

F-measure, F_β , comprehensively considers the importance of precision and recall rate, and gives corresponding weights to precision and recall rate according to specific standards. It provides a balanced evaluation index for the evaluation model. Thus, F_β calculation formula is as follows:

$$F_\beta = (1 + \beta^2) \frac{P * R}{\beta^2 P + R}$$

Where P and R are precision and recall respectively; β^2 parameter, which is set to 0.3.

E-measure, E_ϕ , is used to quantify alignment between predicted and ground truth masks at image and pixel levels. Its calculation formula is as follows:

$$E_\phi = \frac{1}{W * H} \sum_{i=1}^W \sum_{j=1}^H \phi_{FM}(i, j),$$

Where ϕ_{FM} is the enhanced alignment matrix [ho-40].

MAE is used to compute the mean absolute error between predicted and ground truth masks. The calculation formula is as follows:

$$MAE = \frac{1}{W * H} \sum_{i=1}^W \sum_{j=1}^H |T(i, j) - G(i, j)|,$$

Where T and G denote the predicted and ground truth masks, respectively.

Implementation details

In this work, the proposed ConvNeXt-UNet was trained on a GeForce RTX 3090 GPU with 24 GB memory and a pytorch framework. the AdamW optimizer was employed with a learning rate of 0.0001 and weight decay of 0.0004, and the batch size was set to 16. All samples were resized to 352 × 352 before training. An early stopping mechanism was used to supervise the training. Experiments indicated that the model converged to its optimal state at approximately the 120th epoch.

Comparative study

Quantitative analysis

Qualitative analysis

Models	Seen		ClinicDB		Unseen		ColonDB		ETIS		Epoch
	Kvasir-Seg		Dice	iou	CVC-300		Dice	iou	Dice	iou	
HCA-Net	93.6	89.7	95.6	90.9	91.2	87.3	83.4	76.9	83.3	78.9	
Our9388	93.49	87.77	94.92	90.33	90.33	82.52	79.74	69.20	88.69	79.67	100
Ours	92.58	86.41	95.31	91.04	91.49	84.32	83.34	73.75	87.39	77.61	1370
Ours-93.04	03.29				52.19	19.38			84.96	73.85	
Ours-93.66	92.75	86.48	95.20	90.83	91.61	84.52	82.81	72.96	85.39	74.50	2836

Table 1 Quantitative comparison of various models on the Kvasir-SEG dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
CTNet[20-1]	2024	0.917	0.863	0.910	0.928	0.959	0.964	0.023
MEGANet[47-1]	2024	0.913	0.863	0.907	0.918	0.956	0.959	0.025
CAFÉ-Net[2]	2024	0.933	0.889	0.927	0.939	0.967	?	0.019
Polyp-LVT[2]	2024	0.909	0.851	0.913	0.922	0.941	?	0.024
Polyp-Mamba[48-1]	2025	0.919	0.867	0.912	0.951	-	0.968	0.021
MEIN[25-1]	2025	0.905	0.847	0.905	0.914	-	0.964	0.026
MF-Net[2]	2025	0.915	0.863	0.939	0.924	0.962	?	0.022
MSBP-Net[2]	2025	0.919	0.868	0.916	0.926	0.961	?	0.023
CIFFormer[2]	2025	0.926	0.876	0.929	0.939	0.967	?	0.019
PFPRNet[2]	2025	0.930	0.881	0.920	0.935	0.971	?	0.019
PraNet-V2[11-3]	2025	0.907	0.853	?	?	?	?	0.025
MLB-Net[1]	2026	0.926	0.878	0.921	0.938	0.969	0.975	0.021
MSPN[3]	2026	0.927	0.879	?	?	?	?	0.021
DVFIP-Net[4]	2026	0.922	0.874	?	?	?	?	?
Ours	-	0.935	0.879	0.901	0.935	0.952	0.963	0.025

Table 2 Quantitative comparison of various models on the CVC-ClinicDB dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\max \mathbf{E}_\xi$	MAE
CTNet[20-1]	2024	0.936	0.887	0.934	0.952	0.983	0.986	0.006
MEGANet[47-1]	2024	0.938	0.894	0.940	0.950	-	0.987	0.007
Polyp-Mamba[48-1]	2025	0.941	0.896	0.936	0.970	-	0.970	0.008
MEIN[25-1]	2025	0.923	0.878	0.928	0.944	-	0.970	0.008
MF-Net[2]	2025	0.931	0.884	0.953	0.949	0.983	?	0.006

Methods	Year	mDice	mIOU	$\mathbf{F}_{\beta}^w$	$\mathbf{S}_{\alpha}$	$\mathbf{mE}_{\xi}$	$\max \mathbf{E}_{\xi}$	MAE
MSBP-Net[2]	2025	0.940	0.892	0.927	0.939	0.976	?	0.007
CIFFormer[2]	2025	0.944	0.897	0.941	0.934	0.972	?	0.007
PFPRNet[2]	2025	0.949	0.903	0.947	0.957	0.989	?	0.006
PraNet-V2[11-3]	2025	0.929	0.88	?	?	?	?	0.009
MLB-Net[1]	2026	0.945	0.902	0.948	0.958	0.989	0.992	0.006
MSPN[3]	2026	0.943	0.902	?	?	?	?	0.006
DVFIP-Net[4]	2026	0.942	0.896	?	?	?	?	?
Ours	-	0.951	0.906	0.931	0.954	0.983	0.990	0.009

Table 3 Quantitative comparison of various models on the ETIS-LaribPolypDB dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	$\mathbf{F}_{\beta}^w$	$\mathbf{S}_{\alpha}$	$\mathbf{mE}_{\xi}$	$\max \mathbf{E}_{\xi}$	MAE
CTNet[20-1]	2024	0.810	0.734	0.776	0.886	0.913	0.921	0.014
MEGANet[47-1]	2024	0.789	0.709	0.753	0.866	0.855	0.915	0.015
Polyp-Mamba[48-1]	2025	0.756	0.668	0.756	0.887	-	0.872	0.014
MEIN[25-1]	2025	0.788	0.711	0.764	0.870	-	0.902	0.018
MF-Net[2]	2025	0.821	0.750	0.850	0.901	0.925	?	0.011
MSBP-Net[2]	2025	0.795	0.718	0.770	0.873	0.899	?	0.019
CIFFormer[2]	2025	0.810	0.746	0.858	0.889	0.918	?	0.013
PFPRNet[2]	2025	0.828	0.764	0.839	0.908	0.935	?	0.012
PraNet-V2[11-3]	2025	0.713	0.633	?	?	?	?	0.021
MLB-Net[1]	2026	0.821	0.744	0.780	0.899	0.914	0.935	0.013
MSPN[3]	2026	0.798	0.717	?	?	?	?	0.017
DVFIP-Net[4]	2026	0.811	0.730	?	?	?	?	?
Ours	-	0.887	0.797	0.730	0.873	0.883	0.913	0.014

Table 4 Quantitative comparison of various models on the CVC-ColonDB dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	$\mathbf{F}_{\beta}^w$	$\mathbf{S}_{\alpha}$	$\mathbf{mE}_{\xi}$	$\max \mathbf{E}_{\xi}$	MAE
CTNet[20-1]	2024	0.813	0.734	0.801	0.874	0.915	0.919	0.040
MEGANet[47-1]	2024	0.793	0.714	0.779	0.854	0.892	0.895	0.034
Polyp-Mamba[48-1]	2025	0.791	0.713	0.765	0.846	-	0.902	0.034
MEIN[25-1]	2025	0.791	0.715	0.785	0.856	-	0.897	0.034
MF-Net[2]	2025	0.825	0.751	0.864	0.887	0.929	?	0.024
MSBP-Net[2]	2025	0.810	0.731	0.810	0.861	0.912	?	0.029
CIFFormer[2]	2025	0.823	0.741	0.851	0.874	0.918	?	0.026
PFPRNet[2]	2025	0.819	0.741	0.843	0.882	0.915	?	0.028
PraNet-V2[11-3]	2025	0.702	0.620	?	?	?	?	0.044
MLB-Net[1]	2026	0.832	0.757	0.825	0.886	0.924	0.931	0.023
MSPN[3]	2026	0.811	0.734	?	?	?	?	0.029
DVFIP-Net[4]	2026	0.826	0.746	?	?	?	?	?
Ours	-	79.74	69.20	0.759	0.869	0.885	0.903	0.035

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\mathbf{maxE}_\xi$	MAE
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Table 5 Quantitative comparison of various models on the CVC-300(Endoscene) dataset. The best results are highlighted in bold.

Methods	Year	mDice	mIOU	$\mathbf{F}_\beta^w$	$\mathbf{S}_\alpha$	$\mathbf{mE}_\xi$	$\mathbf{maxE}_\xi$	MAE
CTNet[20-1]	2024	0.908	0.844	0.894	0.975	0.975	0.982	0.006
MEGANet[47-1]	2024	0.899	0.834	0.882	0.935	0.966	0.969	0.007
Polyp-Mamba[48-1]	2025	0.906	0.840	0.888	0.936	-	0.975	0.006
MEIN[25-1]	2025	0.893	0.828	0.880	0.930	-	0.965	0.007
MF-Net[2]	2025	0.897	0.836	0.918	0.941	0.966	?	0.007
MSBP-Net[2]	2025	0.903	0.859	0.927	0.942	0.967	?	0.019
CIFFormer[2]	2025	0.883	0.822	0.912	0.939	0.961	?	0.10
PFPRNet[2]	2025	0.891	0.833	0.917	0.928	0.950	?	0.013
PraNet-V2[11-3]	2025	0.873	0.795					0.007
MLB-Net[1]	2026	0.915	0.852	0.909	0.974	0.974	0.978	0.005
MSPN[3]	2026	0.907	0.840					0.006
DVFIP-Net[4]	2026	0.900	0.833	?	?	?	?	?
Ours	-	0.916	0.845	0.848	0.939	0.951	0.969	0.007

Ablation Studies

To systematically evaluate the individual contributions of the proposed modules to the overall segmentation performance, a comprehensive ablation study was conducted. We define our baseline model as the ConvNeXt-Tiny encoder paired with a lightweight decoder and the final prediction head, completely omitting the Multi-Scale Context (MSC) module, BSEI, the Detail Branch (DB), Gated Detail Fusion (GDF), and Deep Supervision (DS). By incrementally adding or systematically removing these components, we isolated their specific impacts on the network. The quantitative results of these architectural modifications are summarized in Table XX.

Effectiveness of MSC

The Multi-Scale Context (MSC) module is designed to capture and aggregate multi-scale contextual information, which is critical for handling significant variations in target object sizes. To assess its impact, we compared the baseline model with a configuration integrating the MSC module. As shown in Table XX, the addition of the MSC module effectively enhanced the network's spatial adaptability, yielding an improvement of XX% in mDice and XX% in mIoU on the XX dataset. This confirms the module's capability to enrich feature representations prior to the decoding stage.

Effectiveness of BSEI

The BSEI module is introduced to [describe specific function, e.g., enhance boundary spatial features / integrate edge-level information] during the feature extraction process. In our ablation experiments, removing the BSEI module from the combined network led to a noticeable degradation in segmentation accuracy, particularly in [describe specific area, e.g., complex boundary regions]. Specifically, the performance metrics decreased by XX% in mDice and XX% in mIoU (Table XX). This demonstrates that the BSEI module successfully contributes to [describe specific benefit, e.g., refining target boundaries and reducing false positives].

Effectiveness of DB

The Detail Branch (DB) was specifically developed to preserve high-resolution spatial details that are typically lost as features pass through the deeper, heavily down-sampled stages of the encoder. Discarding the DB in the ablation experiments resulted in overly smoothed predictions and a loss of fine-grained structural delineation. Table XX reveals that incorporating the DB leads to an XX% gain in mDice on the XX dataset, validating its necessity for extracting and maintaining high-fidelity spatial details throughout the network.

Effectiveness of GDF

The Gated Detail Fusion (GDF) module acts as a selective mechanism to optimally combine the high-resolution spatial features from the DB with the high-level semantic features from the main network. To isolate the contribution of this gating mechanism, we replaced the GDF module with a standard feature fusion strategy, utilizing simple concatenation and element-wise addition. The results in Table XX indicate that the GDF module significantly outperforms simple fusion, achieving an XX% improvement in mIoU. This highlights the GDF's effectiveness in filtering out background noise and dynamically emphasizing salient target features during the fusion process.

Effectiveness of DS

Finally, Deep Supervision (DS) is employed during the training phase to facilitate direct gradient flow to intermediate layers, thereby alleviating the vanishing gradient problem and optimizing multi-level feature learning. When DS was omitted from the complete proposed architecture (evaluating only the final prediction head loss), the model exhibited slower convergence and a drop in final segmentation accuracy. As evidenced in Table XX, the inclusion of DS improved the mDice by XX%, ensuring more robust feature optimization and contributing to the superior generalization of the proposed framework.

Table X. Ablation study of the proposed architectural components on the five test sets.

Dataset	Metrics Params(M) GFlops(G)	Model Baseline	+MSC	+BSEI	+DB	+GDF	+DS
Kvasir-Seg	mDice $\uparrow$						
	mIoU $\uparrow$						
	$\mathbf{F}_{\beta}^w \uparrow$						
	$\mathbf{S}_{\alpha} \uparrow$						
	$\mathbf{mE}_{\xi} \uparrow$						
	$\max \mathbf{E}_{\xi} \uparrow$						
	MAE $\downarrow$						
	mDice $\uparrow$						
CVC-ClinicDB	mIoU $\uparrow$						
	$\mathbf{F}_{\beta}^w \uparrow$						
	$\mathbf{S}_{\alpha} \uparrow$						
	$\mathbf{mE}_{\xi} \uparrow$						
	$\max \mathbf{E}_{\xi} \uparrow$						
	MAE $\downarrow$						
	mDice $\uparrow$						
	mIoU $\uparrow$						
CVC-ColonDB	$\mathbf{F}_{\beta}^w \uparrow$						
	$\mathbf{S}_{\alpha} \uparrow$						
	$\mathbf{mE}_{\xi} \uparrow$						
	$\max \mathbf{E}_{\xi} \uparrow$						
	MAE $\downarrow$						
	mDice $\uparrow$						
	mIoU $\uparrow$						
	$\mathbf{F}_{\beta}^w \uparrow$						
CVC-300	$\mathbf{S}_{\alpha} \uparrow$						
	$\mathbf{mE}_{\xi} \uparrow$						
	$\max \mathbf{E}_{\xi} \uparrow$						
	MAE $\downarrow$						
	mDice $\uparrow$						
	mIoU $\uparrow$						
	$\mathbf{F}_{\beta}^w \uparrow$						
	$\mathbf{S}_{\alpha} \uparrow$						

Dataset	Metrics Params(M) GFlops(G)	Model Baseline	+MSC	+BSEI	+DB	+GDF	+DS
ETIS	mIoU $\uparrow$						
	$\mathbf{F}_{\beta}^w$ $\uparrow$						
	$\mathbf{S}_{\alpha}$ $\uparrow$						
	$\mathbf{mE}_{\xi}$ $\uparrow$						
	$\max \mathbf{E}_{\xi}$ $\uparrow$						
	MAE $\downarrow$						
	mDice $\uparrow$						
	mIoU $\uparrow$						
	$\mathbf{F}_{\beta}^w$ $\uparrow$						
	$\mathbf{S}_{\alpha}$ $\uparrow$						
	$\mathbf{mE}_{\xi}$ $\uparrow$						
	$\max \mathbf{E}_{\xi}$ $\uparrow$						
	MAE $\downarrow$						

Notes. MSC = Multi-Scale Context module; DB = Detail Branch; GDF = Gated Detail Fusion; DS = Deep Supervision. Baseline denotes the ConvNeXt-Tiny encoder with the lightweight decoder and final prediction head, without MSC, BSEI, DB, GDF, or DS.

Discussion

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

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Selected References and Citation Placeholders

The entries below support the current draft structure. Complete and standardize all metadata using the journal's required reference style and a reference manager before submission.

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Methods	Year	Param (M)	GFLOPs (G)
CTNet[20-1]	2024		
MEGANet[47-1]	2024		
Polyp-Mamba[48-1]	2025	49.5	27.9
MEIN[25-1]	2025		
MF-Net[25-2]	2025		
MSBP-Net[34-2]	2025	25.52	12.86
CIFFormer[26-2]	2025	32.44	13.14
PFPRNet[29-2]	2025	118.52	39.39
PraNet-V2[11-3]	2025		
MLB-Net[1]	2026	87.70	
MSPN[3]	2026	28.56	13.21
DVFIP-Net[4]	2026	46.62	19.37
Ours	-	29.60	14.57

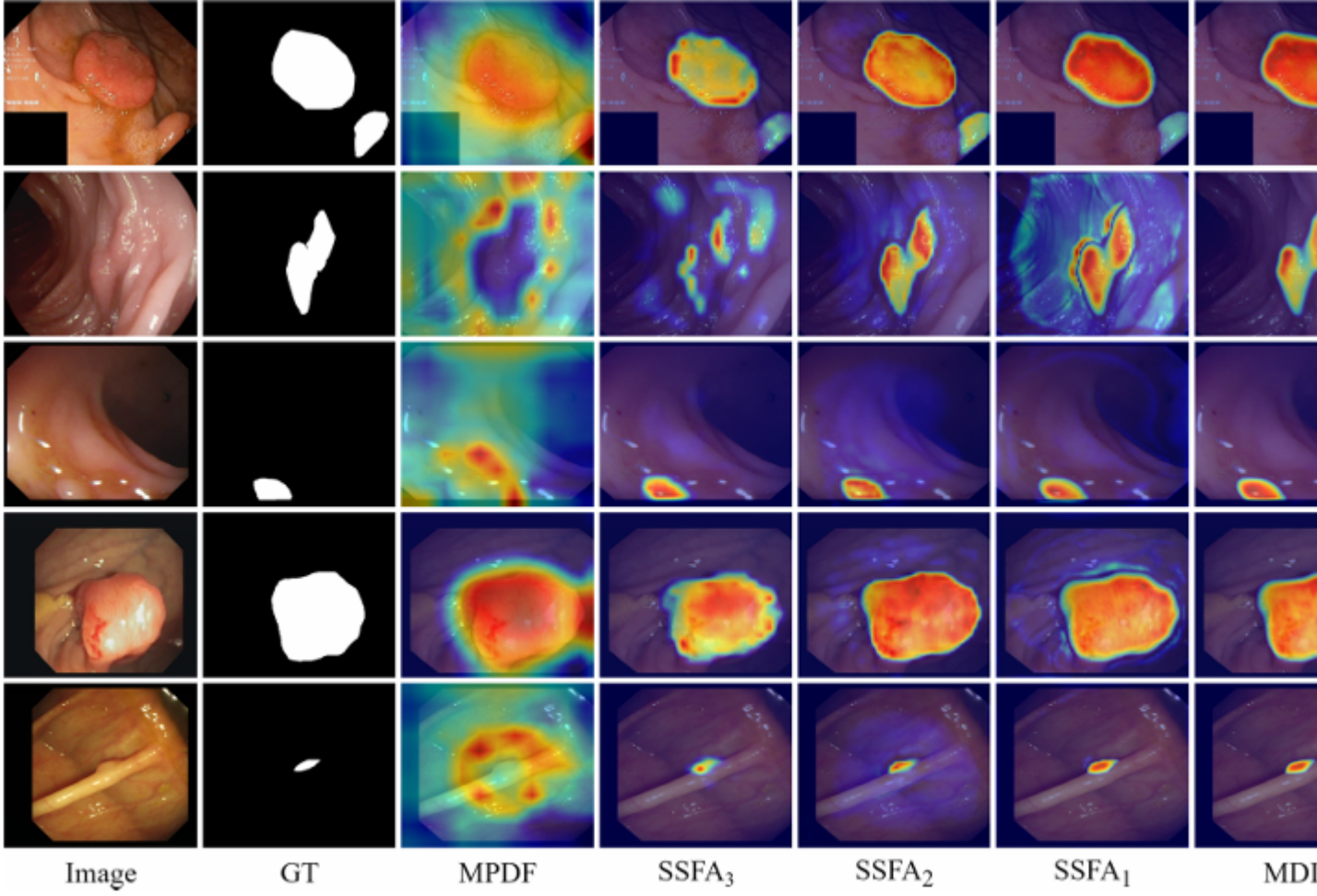


Fig. 6 Attention heatmaps for the model's MPDF, SSFA_i, and MDI modules. Darker colors indicate higher attention

- **MultiScaleContext**: tests whether multi-scale/global context helps.
- **BSEI**: tests boundary-aware feature enhancement.
- **DetailBranch**: tests high-resolution spatial detail recovery.
- **GatedDetailFusion**: tests whether adaptive fusion is better than simple fusion.

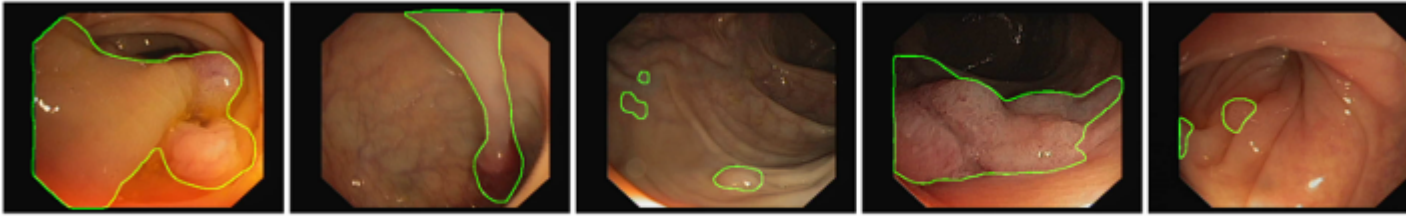


Fig. 1 Six examples of polyp images, with the circled green area in each image representing the polyp

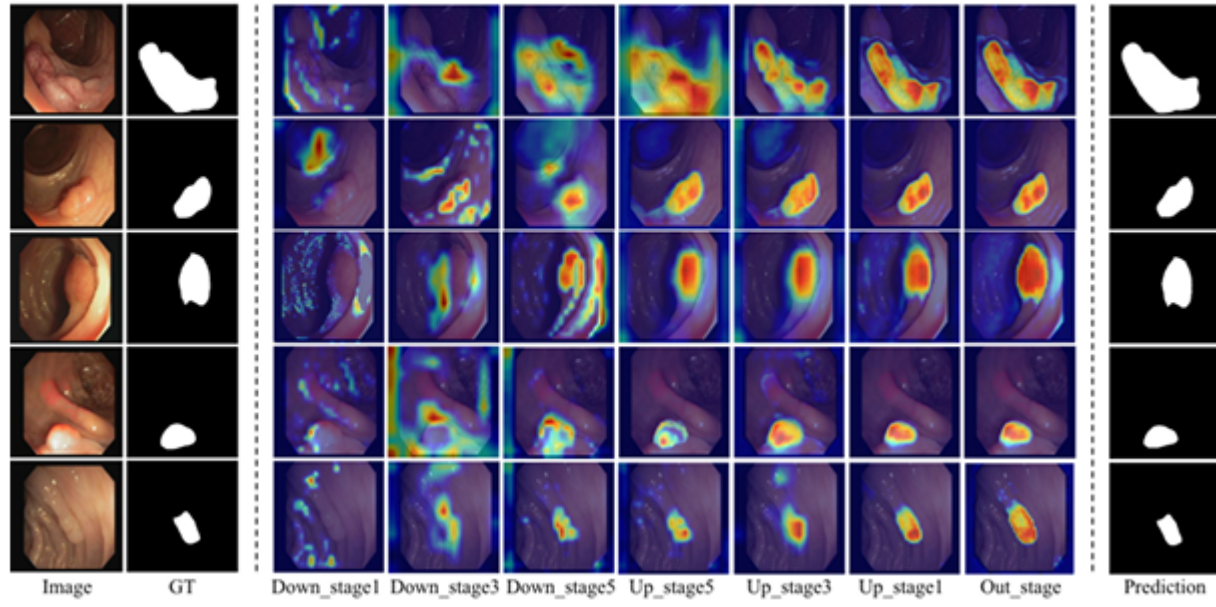


Fig. 8. Module visualization feature map.

TABLE VII
ABLATION STUDY ON BACKBONE WIDTH (BASE_CHANNEL) AND THE COMPUTATIONAL OVERHEAD OF HCA.

base_channel	HCA	CVC-ClinicDB		ETIS-LaribPolypDB		Param (MB)	GFLOPs (G)
		mDice	mIoU	mDice	mIoU		
7	–	0.801	0.748	0.662	0.627	3.45	2.15
7	+	0.902	0.847	0.792	0.742	5.12	3.39
10	–	0.822	0.773	0.693	0.642	6.82	4.05
10	+	0.928	0.879	0.812	0.763	8.94	6.01
12	–	0.843	0.797	0.721	0.672	7.92	6.21
12	+	0.953	0.909	0.833	0.790	10.13	9.33
14	–	0.846	0.802	0.724	0.676	10.77	8.43
14	+	0.956	0.911	0.837	0.794	13.68	12.09
17	–	0.851	0.805	0.729	0.681	15.91	12.11
17	+	0.958	0.913	0.840	0.799	20.36	17.98

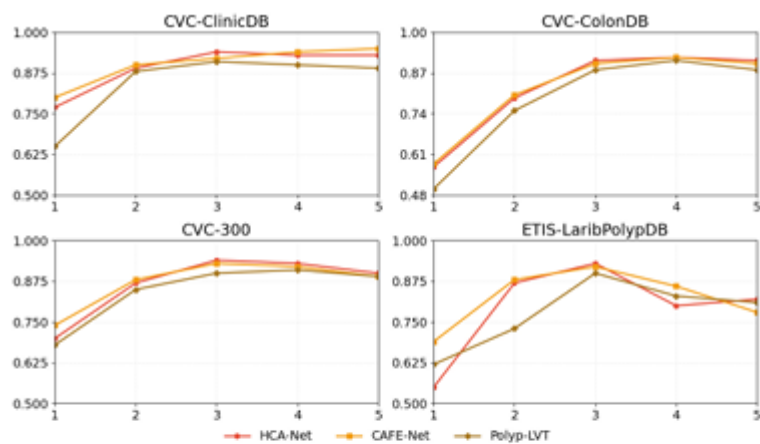


Fig. 6. Performance vs. size on the five polyp datasets. The x-axis denotes the proportion (%) of polyp size, and the y-axis represents the averaged mDice coefficient.

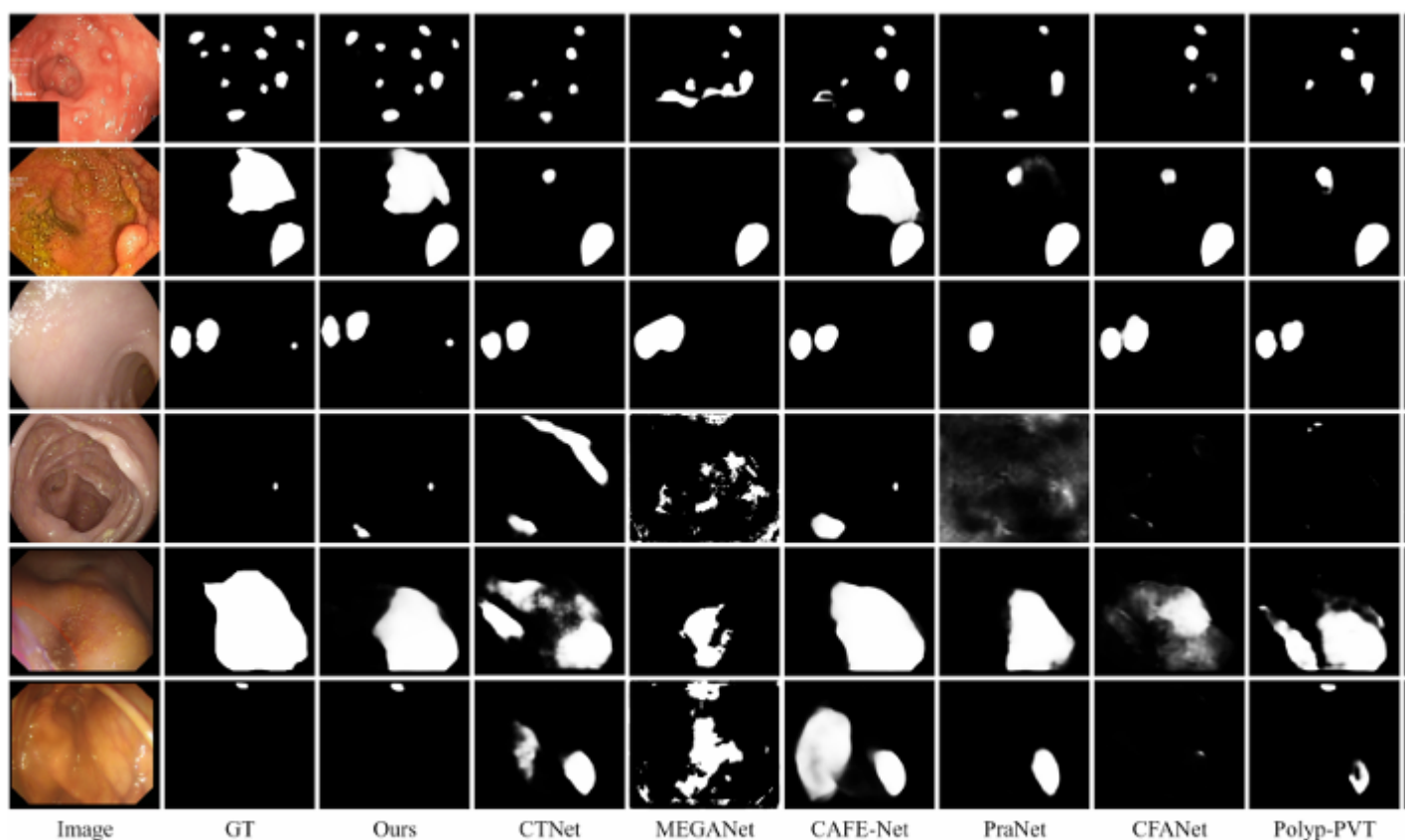


Fig. 4 Our model was compared with the results of various advanced methods on the polyp dataset

- Image(Diagram) of the model with related Modules -> two images
- Heatmap of the model in ablation study with 6 samples
- Heatmap of the model for some stages

- Heatmap of the model in comparison with other models
- Table of the ablation study